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TECHNICAL SUMMARY

Controlled Rectifier Study

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ABSTRACT

Comparing GTO and thyristor rectifiers for isolated micro-grid loads: harmonics, power factor and switching across firing angles. Engineered and compared GTO- and thyristor-based controlled rectifiers for micro-grid applications with isolated loads.

1. Problem and motivation

In a small isolated micro-grid, the harmonics injected by rectifier loads distort the voltage for every other user. The choice of switching device directly affects power quality.

2. Methodology

- Developed MATLAB/Simulink models of both rectifiers and validated their steady-state and transient response
- Investigated harmonic content, power factor variation and switching characteristics across firing angles
- Analyzed trade-offs between the topologies in efficiency and output voltage quality

3. Figures

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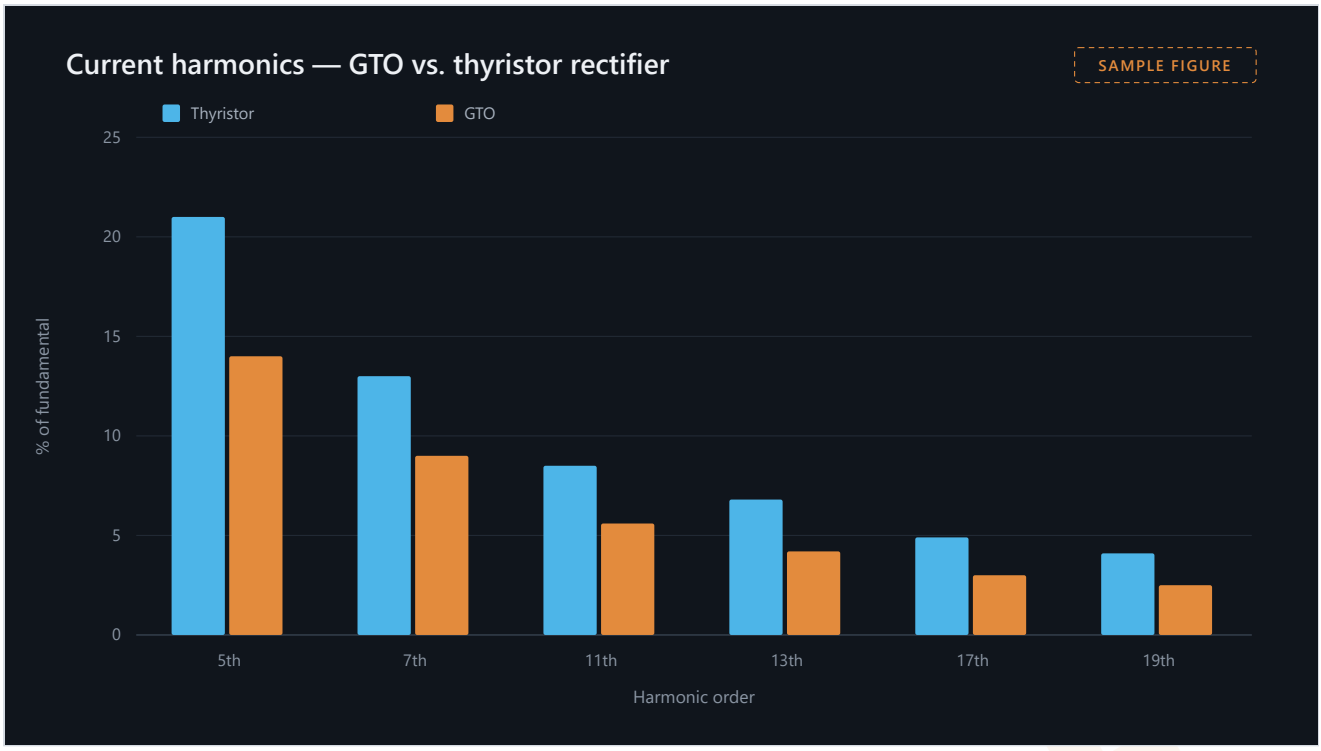


Figure 1. Current harmonics, GTO vs. thyristor rectifier.

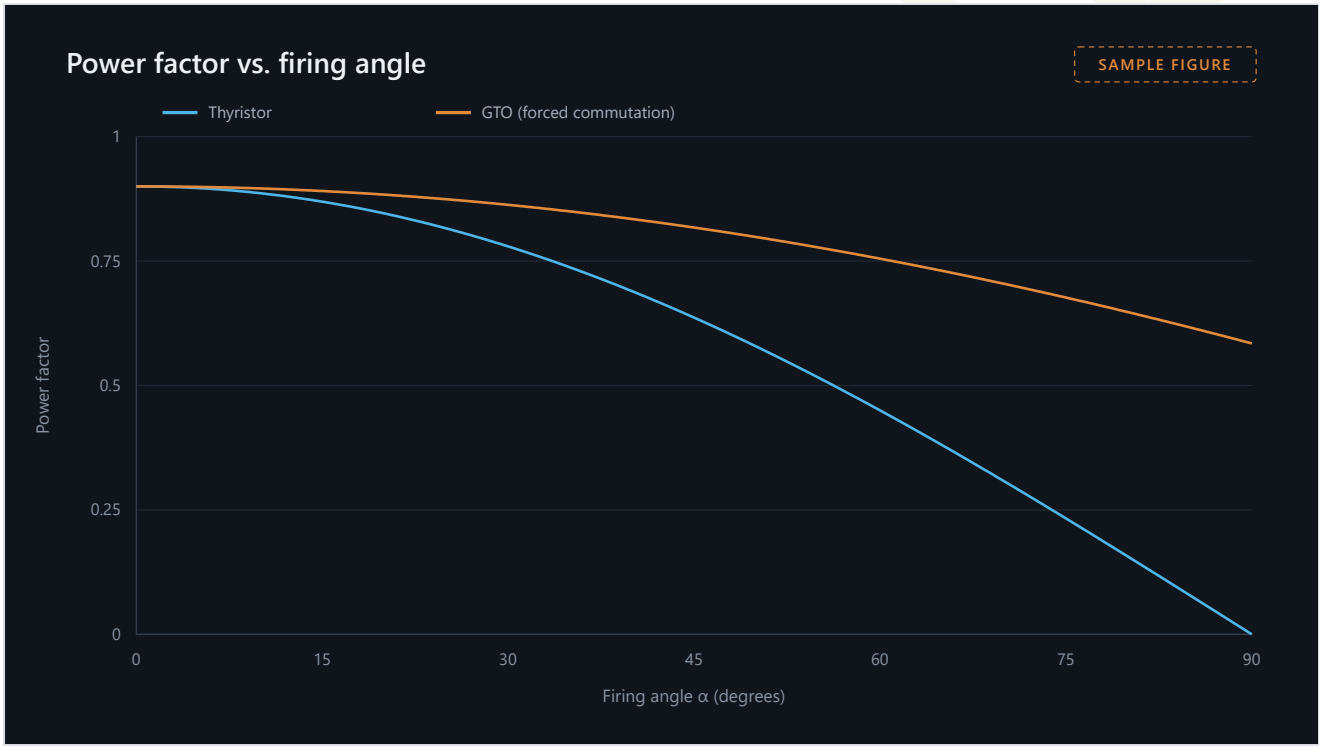


Figure 2. Power factor across firing angles.

4. Team and contribution

Contribution: Built both rectifier models and carried out the comparison.

Tools: MATLAB/Simulink, Proteus

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